PSET 3: Introduction to Matrices and Vectors

Assignment Qs

- Name
- How long did this problem set take you (round to nearest hour)? 0-1 hour 2-3 hours 4-5 hours 5+ hours
- How difficult was this problem set? very easy 1 2 3 4 5 very challenging

Problem 1: Sketch a function

Sketch the graph of a function (any function you like, no need to specify a functional form) that is:

a. Continuous on $[0, 3]$ and has the following properties: an absolute maximum at 0, an absolute minimum at 3, a local maximum at 1 and a local minimum at 2.

Ans.: One valid shape: start high at $x = 0$ (the global max), decrease to a local min somewhere before $x = 1$, rise to a local max at $x = 1$, decrease to a local min at $x = 2$, then continue decreasing down to the global minimum at $x = 3$. Any sketch satisfying these five conditions (endpoints are the global extrema, with one interior local max and one interior local min in between) is acceptable — grade on the qualitative shape, not exact values.

b. Do the same for another function with the following properties: 4 is a critical point (i.e. $f'(x) = 0$ or $f'(x)$ is undefined), but there is no local minimum and no local maximum.

Ans.: Need a horizontal (or vertical/cusp) tangent at $x = 4$ where the function keeps monotonically increasing (or decreasing) through that point — a saddle/inflection point rather than an extremum. A classic example is an S-curve such as $f(x) = (x - 4)^3$ shifted appropriately: $f'(4) = 0$, but f is increasing on both sides of $x = 4$.

Problem 2: Find absolute minimum/maximum values

Find the absolute minimum and absolute maximum values of the functions on the given interval:

a. $f(x) = 3x^2 - 12x + 5,\ [0, 1]$

Ans.:

$$f'(x) = 6x - 12 = 0 \implies x = 2 \quad \text{(outside } [0, 1])$$

So we only need to check the endpoints:

$$f(0) = 5, \qquad f(1) = 3 - 12 + 5 = -4$$

The absolute maximum is 5 at $x = 0$, and the absolute minimum is -4 at $x = 1$.

b. $f(t) = t^2\sqrt{9-t^2},\ [-1,3]$

Ans.:

$$f'(t) = \frac{18t - 3t^3}{\sqrt{9-t^2}} = \frac{3t(6-t^2)}{\sqrt{9-t^2}} = 0 \implies t = 0,\ t = \pm\sqrt{6}$$

Of these, only $t = 0$ and $t = \sqrt{6} \approx 2.449$ lie in $[-1, 3]$. Checking these along with the endpoints:

$$f(-1) = 2\sqrt{2} \approx 2.828, \qquad f(0) = 0, \qquad f(\sqrt{6}) = 6\sqrt{3} \approx 10.392, \qquad f(3) = 0$$

The absolute maximum is $6\sqrt{3} \approx 10.39$ at $t = \sqrt{6}$, and the absolute minimum is 0, achieved at both $t = 0$ and $t = 3$.

c. $s(x) = x - \ln(x),\ [1/2, 2]$

Ans.:

$$s'(x) = 1 - \frac{1}{x} = 0 \implies x = 1$$

Checking the critical point and endpoints:

$$s(1/2) = 0.5 + \ln 2 \approx 1.193, \qquad s(1) = 1, \qquad s(2) = 2 - \ln 2 \approx 1.307$$

The absolute minimum is 1 at $x = 1$, and the absolute maximum is $2 - \ln 2 \approx 1.307$ at $x = 2$.

Problem 3: Second derivatives, concavity, and inflection

$$f(x) = \frac{x^3}{15} - x^2 + 4x + 8$$

a. Find $f'(x)$.

Ans.: $f'(x) = \dfrac{x^2}{5} - 2x + 4$

b. Find $f''(x)$.

Ans.: $f''(x) = \dfrac{2x}{5} - 2$

c. Find $f'''(x)$.

Ans.: $f'''(x) = \dfrac{2}{5}$

d. Find all critical points of f and classify each as a local maximum or local minimum using the second derivative test.

Ans.: Setting $f'(x) = 0$:

$$\frac{x^2}{5} - 2x + 4 = 0 \implies x^2 - 10x + 20 = 0 \implies x = 5 \pm \sqrt{5}$$

Applying the second derivative test:

- At $x = 5 - \sqrt{5} \approx 2.764$: $f''(x) \approx -0.894 < 0 \implies$ local maximum
- At $x = 5 + \sqrt{5} \approx 7.236$: $f''(x) \approx +0.894 > 0 \implies$ local minimum

e. Find the inflection point, and state the intervals on which f is concave up and concave down.

Ans.: Setting $f''(x) = 0$:

$$\frac{2x}{5} - 2 = 0 \implies x = 5$$

$$f(5) = \frac{125}{15} - 25 + 20 + 8 = \frac{25}{3} + 3 = \frac{34}{3} \approx 11.33$$

The inflection point is $\left(5,\ \frac{34}{3}\right)$. Since $f''(x) < 0$ for $x < 5$ and $f''(x) > 0$ for $x > 5$, f is concave down on $(-\infty, 5)$ and concave up on $(5, \infty)$.

Problem 4: Vector arithmetic

Suppose

$$\mathbf{x} = \begin{pmatrix} 3 \\ 2q \\ 4 \end{pmatrix} \quad \text{and} \quad \mathbf{y} = \begin{pmatrix} p+2 \\ -5 \\ 3r \end{pmatrix}$$

If $\mathbf{x} = 2\mathbf{y}$, find p, q, r.

Ans.: Matching components of $\mathbf{x} = 2\mathbf{y}$:

$$3 = 2(p+2) \implies 3 = 2p + 4 \implies p = -\frac{1}{2}$$

$$2q = 2(-5) \implies 2q = -10 \implies q = -5$$

$$4 = 2(3r) \implies 4 = 6r \implies r = \frac{2}{3}$$

So $p = -\frac{1}{2},\ q = -5,\ r = \frac{2}{3}$.

Problem 5: Check for linear dependence

Which of the following sets of vectors are linearly dependent?

a. $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}, \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix}$

Ans.:

$$\det \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix} = 1(45 - 48) - 4(18 - 24) + 7(12 - 15) = -3 + 24 - 21 = 0$$

Since the determinant is 0, these vectors are **linearly dependent**. Indeed, $2(4, 5, 6) - (1, 2, 3) = (7, 8, 9)$.

b. $\begin{pmatrix} 13 \\ 7 \\ 9 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ -2 \\ 5 \\ 8 \end{pmatrix}$

Ans.: This set contains the zero vector, so it is automatically **linearly dependent**: $0 \cdot a + 1 \cdot b + 0 \cdot c = \mathbf{0}$ is a nontrivial combination that sums to zero.

c. $\begin{pmatrix}1\\2\\1\end{pmatrix}, \begin{pmatrix}2\\-2\\-1\end{pmatrix}, \begin{pmatrix}2\\1\\3\end{pmatrix}$

Ans.:

$$\det\begin{pmatrix}1 & 2 & 2\\2 & -2 & 1\\1 & -1 & 3\end{pmatrix} = 1(-6+1) - 2(6-1) + 2(-2+2) = -5 - 10 + 0 = -15$$

Since the determinant is nonzero, these vectors are **linearly independent**.

Problem 6: Vector length

Find the length of the following vectors:

a. $(4, 2)$

Ans.: $\sqrt{16+4} = \sqrt{20} = 2\sqrt{5} \approx 4.47$

b. $(4, 2, 3, 1)$

Ans.: $\sqrt{16+4+9+1} = \sqrt{30} \approx 5.48$

c. $(0, 0, 0, 0, 3)$

Ans.: $\sqrt{9} = 3$

Problem 7: Documents

Three documents are represented by their counts of the words (tax, vote, reform, budget):

$$\text{Doc1} = (1, 1, 3, 5) \qquad \text{Doc2} = (2, 0, 0, 1) \qquad \text{Doc3} = (3, 0, 1, 2)$$

a. Compute the cosine similarity of Doc1 with Doc2, and of Doc1 with Doc3. Which document is more similar to Doc1?

Ans.: $\|\text{Doc1}\| = 6$, $\|\text{Doc2}\| = \sqrt{5}$, $\|\text{Doc3}\| = \sqrt{14}$.

$$\cos(\text{Doc1}, \text{Doc2}) = \frac{7}{6\sqrt{5}} \approx 0.522, \qquad \cos(\text{Doc1}, \text{Doc3}) = \frac{16}{6\sqrt{14}} \approx 0.713$$

Since $0.713 > 0.522$, **Doc3** is more similar to Doc1.

b. Suppose Doc2 were twice as long, with every word count doubled: $\text{Doc2}^* = (4, 0, 0, 2)$. Compute the cosine similarity of Doc1 with Doc2^*. Compare to your answer above and explain why this is a desirable property of the measure.

Ans.: $\|\text{Doc2}^*\| = \sqrt{20}$:

$$\cos(\text{Doc1}, \text{Doc2}^*) = \frac{14}{6\sqrt{20}} \approx 0.522$$

This is identical to $\cos(\text{Doc1}, \text{Doc2})$ from part (a). Cosine similarity normalizes by vector length, so it measures the angle between vectors — i.e. relative word proportions — rather than raw magnitude. This means documents of different lengths that discuss the same topics in the same proportions are judged equally similar, which is exactly the desired behavior when comparing documents of different lengths.

Problem 8: Matrix algebra

Using the matrices below, calculate the following. Some may not be defined; if that is the case, say so.

$$A = \begin{pmatrix} 3 \\ -2 \\ 9 \end{pmatrix} \quad B = \begin{pmatrix} 8 \\ 0 \\ -1 \end{pmatrix} \quad C = \begin{pmatrix} 7 & -1 & 5 \\ 0 & 2 & -4 \end{pmatrix} \quad D = \begin{pmatrix} 3 & 1 \\ 3 & 4 \\ 3 & -7 \end{pmatrix}$$

$$E = \begin{pmatrix} 5 & 2 & 3 \\ 1 & 0 & -4 \\ -2 & 1 & -6 \end{pmatrix} \quad F = \begin{pmatrix} 4 & 1 & -5 \\ 0 & 7 & 7 \\ 2 & -3 & 0 \end{pmatrix} \quad G = \begin{pmatrix} 2 & -8 & -5 \\ -3 & 7 & -4 \\ 1 & 0 & 3 \\ 1 & 2 & 6 \end{pmatrix}$$

$$K = \begin{pmatrix} 9 \\ -2 \\ -1 \\ 0 \end{pmatrix} \qquad L = \begin{pmatrix} 5 & 0 & 3 & 1 \end{pmatrix}$$

a. $A + B$

Ans.: $\begin{pmatrix} 3+8 \\ -2+0 \\ 9-1 \end{pmatrix} = \begin{pmatrix} 11 \\ -2 \\ 8 \end{pmatrix}$

b. $A'B$

Ans.: A' is 1×3 and B is 3×1, so $A'B$ is defined and is a scalar:

$$A'B = (3)(8) + (-2)(0) + (9)(-1) = 24 + 0 - 9 = 15$$

c. $C + D$

Ans.: C is 2×3 and D is 3×2 — dimensions do not match, so $C + D$ is **not defined**.

d. LK

Ans.: L is 1×4 and K is 4×1, so LK is defined and is a scalar:

$$LK = (5)(9) + (0)(-2) + (3)(-1) + (1)(0) = 45 + 0 - 3 + 0 = 42$$

e. KL

Ans.: K is 4×1 and L is 1×4, so KL is a 4×4 matrix:

$$KL = \begin{pmatrix} 45 & 0 & 27 & 9 \\ -10 & 0 & -6 & -2 \\ -5 & 0 & -3 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

f. $E - 5I_3$

Ans.:

$$E - 5I_3 = \begin{pmatrix} 5 & 2 & 3 \\ 1 & 0 & -4 \\ -2 & 1 & -6 \end{pmatrix} - \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix} = \begin{pmatrix} 0 & 2 & 3 \\ 1 & -5 & -4 \\ -2 & 1 & -11 \end{pmatrix}$$

g. BC

Ans.: B is 3×1 and C is 2×3. The inner dimensions (1 and 2) do not match, so BC is **not defined**.

AI and Resources statement

Please list (in detail) all resources you used for this assignment. If you worked with people, list them here as well. It is not enough to say that you used a resource for help, you need to be specific on the link and how it was helpful. W/R/T gen AI tools (including GPT, Claude, etc.) you cannot use them to do work on your behalf—you cannot put in any of the questions, etc. You can ask for help on logic / sample problems. If you do use GPT or other AI tools, you need to provide a link to your chat transcript. Any suspected academic integrity violations will be immediately reported to the Dean of Students.